



Department of Physical Sciences
Faculty of Applied Sciences
Rajarata University of Sri Lanka, Mihintale
B.Sc. in Applied Sciences

STA 1303 – Probability Theory
Tutorial No: 4

1. The average age at first marriage is 25 for women and 27.8 for men in the US. If the standard deviation for women is four years, what is the probability that a random selection of 32 women will have an average age at first marriage between 26 and 27?
2. In a communication system each data packet consists of 1000 bits. Due to the noise, each bit may be received in error with probability 0.1. it is assumed bit errors occur independently. Find the probability that there are more than 120 errors in certain data packet.
3. Achievement test scores of all high school seniors in a state have mean 60 and variance 64. A random sample of $n = 100$ students from one large high school had a mean score of 58. Is there evidence to suggest that this high school is inferior? (Calculate the probability that the sample mean is at most 58 when $n = 100$.)
4. The service times for customers coming through a checkout counter in a retail store are independent random variables with mean 1.5 minutes and variance 1.0. Approximate the probability that 100 customers can be served in less than 2 hours of total service time
5. you have invited 64 guests to party. You need to make sandwiches for the guest. You believe that guest might need 0, 1, or 2 sandwiches with probabilities, $\frac{1}{4}$, $\frac{1}{2}$, and $\frac{1}{4}$ respectively. You assume that the number of sandwiches each guest needs is independent from other guest. How many sandwiches should you make so that you are 95% sure that no shortage?
6. An anthropologist wishes to estimate the average height of men for a certain race of people. If the population standard deviation is assumed to be 2.5 inches and if she randomly samples 100 men, find the probability that the difference between the sample mean and the true population mean will not exceed .5 inch

(Please answer all the questions and submit the answer script to the tutorial room on or 01ST of November 2023)